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Explainable Anomaly Detection for Grid Monitoring using Probabilistic Load Forecasting

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***Abstract*—Anomaly detection (AD) plays a crucial role in smart grids. The integration of power-intensive, internet-connected technologies in the residential sector gives rise to new risk scenarios, which can compromise grid stability if not promptly detected. Forecasting-based AD offers a real-time capable solution for grid operators and provides interpretable results. However, commonly used deterministic threshold-based techniques struggle with heavily fluctuating data at high temporal resolution. In this paper, we propose a novel probabilistic real-time AD framework. The two-stage approach combines probabilistic very-short-term load forecasting with a quantile-based scoring function. By considering the predictive distribution of power demand based on external features, our method characterizes the inherent uncertainty in power demand patterns, allowing for clearer distinction between normal variations and critical anomalies. Our method is tested on real aggregate household data, from which we can detect abnormal demand patterns of individual households. In addition, our method is evaluated with synthetic anomalies using classification metrics and the receiver-operator characteristic (ROC), demonstrating superior performance over conventional deterministic methods. By leveraging probabilistic forecasting for AD, our method offers a straightforward, transparent, yet accurate and robust tool for smart grid nowcasting. The proposed real-time framework can provide grid operators with a warning system to anticipate and mitigate critical situations timely, thereby enhancing power system resilience.**

***Index Terms*—Anomaly Detection, Probabilistic Load Forecasting, Quantile Regression, Grid Monitoring, Resilience**

I. Introduction

THE residential sector accounts for around one third of total end-use electricity demand, e.g. 37% in the U.S. [1]. With the ongoing electrification, driven by emerging technologies such as electric vehicles (EVs), heat pumps (HPs), and air conditioning (AC), household electricity consumption is expected to rise. These energy-intensive technologies introduce significant load fluctuations that can challenge grid stability. Additionally, the increased use of information and communication technology in distribution systems, e.g. demand-side management (DSM), creates new risks scenarios, for example, synchronization and rebound effects [2] or load altering attacks [3]. Synchronized power demands can have significant and abrupt impacts on the aggregated load [4], which can affect grid stability, emphasized by real-world events of collective human activities [5], [6]. Therefore, it is crucial to detect abnormal consumption patterns or attacks that necessitate attention or intervention by grid operators in time.

Anomaly detection (AD) in power systems has various applications, e.g., improving energy efficiency, fault detection, detecting cyber attacks, or combating electricity theft [7], [8]. However, the monitoring of household power consumption in real-time frameworks, such as digital twins [9], has received little attention yet. Unlike cyber attacks targeting communication networks, consumption-based anomalies must be identified through load pattern analysis, as they bypass conventional security monitoring systems. AD in the active load is challenging, especially in aggregated form, as methods must distinguish between typical statistical variations in power demand and critical deviations indicative of anomalous behavior. In the context of grid monitoring for resilience, AD systems must satisfy several key requirements: real-time capability, transparency for user interpretability, high accuracy with minimal false alarms, no reliance on the availability of smart meter data, and the ability to recognize unprecedented anomalies without labeled training data.

The literature on AD methods is dominated by complex machine learning (ML), particularly deep learning (DL) models [7], [8]. While these techniques can achieve high performance in AD, their applicability in grid monitoring is limited. First, they require extensive training data, which is often unavailable for anomalies, especially in labeled form [7]. Second, the black-box nature of DL models hampers explainability, making it difficult for domain experts to contextualize anomalies. This lack of transparency in safety-critical environments raises concerns for compliance with regulatory frameworks for critical infrastructures like the EU AI Act [10]. As interpretability and trustworthiness remain fundamental, unsolved issues of DL methods [7], [8], there is a need for explainable models that provide reasoning for why a power consumption event is flagged as abnormal.

Forecasting-based approaches present a promising alternative, as they ground on load predictions to identify anomalies by comparing expected and actual power consumption. In this two-stage approach, they can simultaneously perform electrical

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load forecasting (LF) and unsupervised AD in real-time [11]–[13]. By building on existing grid operation practices and not requiring labeled data, this approach provides a straightforward and transparent framework for grid monitoring. However, most studies employing this approach use deterministic point forecasts, typically combined with fixed thresholds such as the "two-sigma rule" [14]. This results in suboptimal performance, particularly in high-resolution data with short-term fluctuations, where absolute thresholds can lead to low accuracy, undermining reliability in practical applications.

In this work, we propose a novel probabilistic forecasting-based real-time AD framework. Our approach combines probabilistic very-short-term LF with a quantile-scoring technique. In contrast to previous LF-based AD methods relying on point forecasts and fixed thresholds, our method introduces an anomaly score by weighting deviations in consumption data according to the load prediction quantile. This quantification of uncertainty helps effectively distinguishing between usual variations and critical deviations. The anomaly score is further refined using an exponential smoothing filter to enhance the detection of collective anomalies over time.

For the LF model, we use an auto-regressive moving average approach with exogenous variables (ARMAX) and quantile regression (QR), a technique proven effective for AD in time-series data [13], [15]. We compare QR-based forecasting with a gradient boosting approach (XGBoost [16]) to assess the trade-off between transparency and forecasting accuracy.

We evaluate our method using real household power consumption data in 15-minute resolution from [17]. From the aggregated demand of 21 households, we can detect plausibly anomalous load patterns created by single households. Moreover, we conduct a systematic analysis for benchmarking classification performance using synthetic anomaly scenarios and analyze the receiver operating characteristic (ROC). We demonstrate that our method significantly outperforms traditional deterministic approaches, reducing false positives while maintaining high recall rates. Additionally, we found that even though XGBoost achieves better forecasting metrics, QR delivers comparable detection performance.

Our study demonstrates the potential of integrating probabilistic LF into explainable real-time AD for grid monitoring. In contrast to the trend of creating highly accurate forecasts with complex DL techniques, we show that the probabilistic AD method can deal effectively with load fluctuations, even with simple LF models like QR/ARMAX, providing greater interpretability. The anomaly score, avoiding binary classification, can serve as an intuitive warning indicator for grid operators to detect abnormal situations before they endanger grid stability and lead to critical damage, thus increasing response time and grid resilience. By requiring only aggregated load measurements, e.g. from secondary substations, our method can be readily implemented in a non-invasive way. The code used for this study is made freely available[1].

[1]https://github.com/EINS-TUDa/EXPLAD-Anomaly-Detection-with-Probabilistic-Forecasting

The rest of the paper is structured as follows. Section II provides an overview of related work on AD in power systems. Section III explains the methodology of our proposed method, which is evaluated in Section IV. A discussion on the approach and results is provided in Section V. We conclude in Section VI.

II. Related Work

In this section, we review the related work in the field of AD for power systems, with a special focus on load anomalies and forecasting-based approaches. In our overview, we aim to clearly differentiate between methods used for LF and the kind of AD itself, which is often unconsidered in existing reviews.

Typical use cases of AD in power systems concerned applications like identifying non-technical losses such as electricity theft [18], fault detection [19], false data injection in system state estimation [20], and building energy consumption efficiency [11], [21], [22]. The real-time monitoring of critical consumer behavior for grid stability appears to lack consideration in the literature.

AD methods can be divided into supervised and unsupervised methods [7]. Among supervised methods, prior work has considered support vector machines (SVMs), convolutional neural networks (CNNs), or random forests (RFs) [7]. As a major drawback, supervised learning relies on large labeled datasets, which are often nonexistent or not accessible and may be insufficient to fully characterize all notions of anomalousness. Moreover, due to the nature of anomalies, abnormal data occurs much less frequent than normal data, leading to the problem of imbalanced classes [7]. Given these limitations, the use of supervised methods is restricted to specific use-cases where labeled data is available and the type of anomaly can be clearly specified, e.g. identifying electricity theft.

In light of the mentioned issues, unsupervised constitutes the predominant approach to AD [8]. Here, the central idea is to learn a model of normality from normal data in an unsupervised manner so that anomalies become detectable through deviations from the model. This deviation can be expressed in the form of an *anomaly score*, providing a continuous measure quantifying the degree of abnormality. In this way, no labeled anomaly

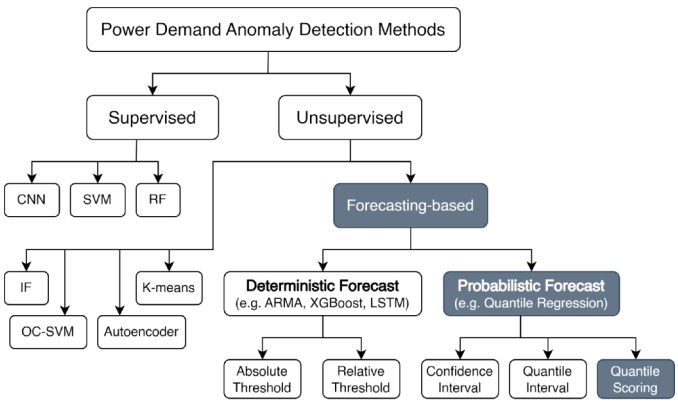


Fig. 1. Categorization of methods used for anomaly detection in power demand

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data is required for training and even unprecedented anomalies can be detected.

As shown in Fig. 1, we further separate unsupervised AD methods into two primary methodological approaches: purely data-driven and forecasting-based methods. Several generic ML algorithms have been applied for AD, including one class SVM (OC-SVM), Isolation Forest (IF), distance-based methods such as K-means clustering, or density-based methods like local outlier factor (LOF). Moreover, DL techniques, such as auto-encoders, gain popularity [8]. However, many of these data-driven ML techniques are sensitive to noise, prone to over-fitting, and lack explainability. They operate directly on the raw data, without considering the electrical load as a temporal-sequential, context-specific process. In addition, they might be difficult to apply in a real-time context, as they usually process a whole dataset at once [14].

In contrast, the forecasting-based approach takes a domain-specific perspective and resembles a two-stage approach by using LF to predict the consumption and interpret deviations as indicator for anomalies. This approach can be efficiently integrated in a real-time framework and has emerged suitable for marking anomalies in a transparent way [11]–[13].

For the LF model, there exist a multitude of proposed methods, in turn. Point forecast models include statistical approaches like (auto-)regression [11], [12], gradient boosting machine techniques, random forest [23], and DL techniques such as long short-term memory (LSTM) [24], [25],

The detection of anomalies is usually performed by analyzing the residuals based on the current load value in respect to the forecasted power consumption. For deterministic predictions, most commonly a fixed threshold like the 2- (or 3-) sigma rule is used [11], [14], [23], [25], defining an anomaly as two standard deviations above or below the predicted value. This rule is based on the assumption of normally distributed prediction errors, which is often invalid, leading to poor accuracy. Moreover, absolute thresholds like the 2-sigma rule struggle with heteroskedasticity and short-term fluctuations, which appear at high temporal resolutions, e.g. 15 minute.

As deterministic LF cannot adequately capture the stochastic variability inherent in power demand patterns, probabilistic load forecasting (PLF) may fill this gap. By providing confidence intervals or a full distribution rather than point estimates, PLF methods can characterize power demand more accurately and quantify the uncertainty in the prediction [26]. Despite its potential, probabilistic approaches have seen limited application for AD yet. Tambuwal and Neagu [15] proposed to use the quantile interval based on QR as anomaly indicator for AD in time-series, considering different application domains, demonstrating its effective performance over methods that assumed a Gaussian prediction distribution. Beykirch et al. [27] considered a probabilistic forecast combination for AD in building level heat load time-series, using fixed quantile thresholds for the lower and upper bound. Stjelja et al. [22] considered probabilistic forecasting for electrical and heating energy consumption AD, using fixed quantile bounds for binary classification as well. While the use of quantile thresholds instead of absolute deviations from a point forecast can improve AD performance, none of these works has considered using the complete non-parametric PLF distribution for quantifying the level of anomalousness in the form of a continuous anomaly score.

III. Probabilistic Anomaly Detection Method

This section details our proposed method for real-time power demand AD. The method is split into two main parts. First, a PLF model forecasts the expected power demand based on exogenous variables. Second, the real-time power consumption measurement is compared against the forecast to compute an anomaly score. This score serves as a warning indicator for monitoring and can optionally be thresholded for binary classification. An overview of the approach is illustrated in Fig. 2.

The two parts of the approach – forecast and detection – are modular and independent, thus can be exchanged with other models. The probabilistic load forecast serves as the interface between the stages.

A. Probabilistic Load Forecasting

The PLF model outputs a probability distribution describing the estimated load and its associated uncertainty for each time step. External features and past load values constitute the input of the PLF model. Two PLF models are used for comparison: Linear QR and quantile-based XGBoost.

1) Features: For both forecast models, we use auto-regressive moving average and exogenous variables (ARMAX) as features. The exogenous variables include calendar information, such as time of day, day of week, month of year, and weather variables, including temperature, solar irradiation, and humidity. As auto-regressive variables, we use lagged power consumption values as one-day ahead forecast. Some of the variables are incorporated as moving averages, which helps to deal with short-term fluctuations. The selection of feature variables was manually optimized. Beside the ARMAX features, we also consider a model utilizing only exogenous variables without auto-regressive inputs for comparison.

2) Quantile Regression: QR is based on simple linear regression, using the features x_j as a linear combination

$$\hat{y_t} = \beta_0 + \sum_{j=1}^{m} \beta_j x_{j,t} \tag{1}$$

where $\hat{y_t}$ represents the load at time step t. Instead of minimizing the mean squared error like in classical linear regression, QR minimizes the pinball loss, defined as

$$L_q(y_t, \hat{y_t}) = \begin{cases} q(y_t - \hat{y_t}) & \text{if } y_t \geq \hat{y_t}, \\ (1-q)(\hat{y_t} - y_t) & \text{if } y_t < \hat{y_t}. \end{cases} \tag{2}$$

where $q \in [0, 1]$ represents a given target quantile.

By computing predictions for several quantiles, a non-parametric probability distribution is obtained, avoiding the need for restrictive assumptions on the underlying data distribution. This distribution characterizes the uncertainty inherent in the prediction.

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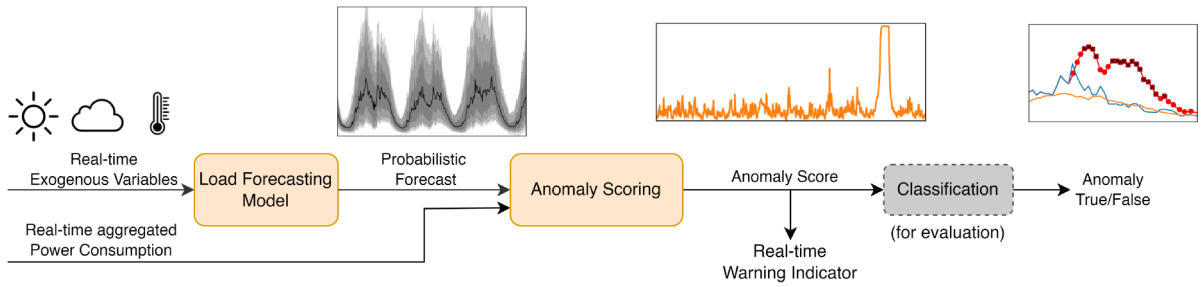


Fig. 2. Overview of the approach for real-time power demand AD

3) XGBoost: XGBoost [16] is an optimized gradient boosting framework that constructs an ensemble of decision trees for regression tasks. Gradient boosting machine techniques are particularly effective for handling complex, non-linear dependencies in time-series forecasting. As with QR, the loss function can be adjusted such that quantile predictions can be obtained. In similar vein, we use XGBoost as a non-linear alternative for comparing its AD performance.

B. Anomaly Scoring

The anomaly scoring (AS) mechanism assesses the degree of abnormality for a given real-time measurement by comparing it with the probabilistic forecast. This is achieved by determining the quantile interval in which the actual measurement falls, similar as in [15]. For accurate quantile estimation, a high resolution is important, especially near the extreme quantiles (0 and 1). Since the forecast distribution consists of discrete quantiles, linear interpolation is applied to determine the quantile q_t corresponding to the observed measurement y_t more accurately. The prediction quantile q_t characterizes the prediction error as the level of (un)expectation of the observed load y_t with respect to the probabilistic forecast.

The anomaly score is defined as a function of the quantile q_t in a way that values near the median (0.5) result in a low score, indicating normal situations, whereas values near the extremes (0 and 1) yield a high score, indicating abnormal behavior. For this purpose, a U-shaped function on the interval $[0, 1]$ is employed for weighting values near the extreme quantiles accordingly. We use the *Logit* function as

$$AS_t = \left| \ln \frac{q_t}{1 - q_t} \right| \tag{3}$$

for a given quantile q_t at time step t. This function is symmetric around 0.5, i.e. positive and negative outliers are weighted equally regarding their quantile.

To improve robustness and consider the sequential nature of anomalies, an exponential smoothing filter is applied to the anomaly score:

$$AS_t^{\text{smoothed}} = \alpha AS_t + (1 - \alpha) AS_{t-1} \tag{4}$$

where $\alpha \in [0, 1]$ is the smoothing parameter and set to $\alpha = 0.5$. This filtering process helps to distinguish between transient outliers and sustained anomalies, improving detection of collective anomalies that persist over multiple time steps (e.g. several hours).

IV. Evaluation

We evaluate our proposed method using real-world high-resolution household consumption data. After introducing the dataset, the performance of the LF models is analyzed and compared. We then present real anomalous situations found in the dataset, followed by a comparative benchmark analysis.

A. Dataset

We use the dataset [17] which contains the load profiles of single-family households in Germany and weather data. After identifying and removing households with PV or incomplete data, to obtain a consistent data basis, 21 households remain. Their aggregated load (without heat pumps) of the whole years 2019 and 2020 is used in 15-minute resolution, whereas the data from 2019 is used for training the forecast model and the 2020 data is used for testing. The individual data is only used for validating discovered abnormal situations.

B. Load Forecasting

After training the forecast models on the aggregated load and weather data from 2019, their (median) predictions on the testing period (2020) are evaluated using the coefficient of determination and mean absolute percentage error (MAPE) $\frac{1}{n}\sum_{t=1}^{n}\left|\frac{\hat{y}_t - y_t}{y_t}\right|$. The results are summarized in Table I. In general, XGBoost achieves better metrics than QR. Furthermore, adding auto-regressive variables improves performance, especially for QR.

TABLE I
Forecast Models Performance

Features	Model	R^2 (%)	MAPE (%)
Auto-regr. + Ex. var.	Quantile Regression	61.26	15.81
	Q-XGBoost	62.66	15.21
Ex. var. only	Quantile Regression	51.08	18.38
	Q-XGBoost	59.59	15.81

C. Real Anomalies in Dataset

After applying our AD method, we discovered multiple clearly abnormal patterns in the dataset. For example, we could detect a temporarily increased aggregated load, tracing back to presumably corrupted data of household #35 due to a communication error in the data recording system: On 9th August 2020, the household's load appears typical until 9:56

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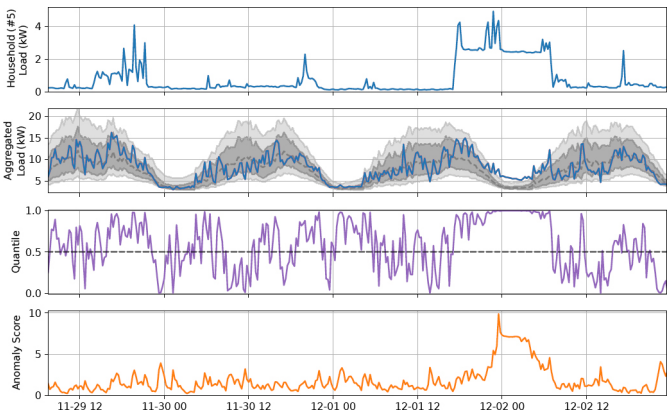


Fig. 3. Detection of abnormal consumption pattern (individual load of household #5) with the smoothed anomaly score. The second plot depicts the observed aggregated load (blue) and a corresponding forecast based on QR/ARMAX, showing median (dashed) and the 10% and 1% quantile intervals. The third plot shows the prediction quantile in which the observed load falls.

a.m., when it quickly reaches 4639.82 W, where it stays exactly constant until 11th August 9:56 a.m. After that, only NaN values are present for this household in the dataset.

Fig. 3 demonstrates the detection of another real anomalous situation, occurred during the night of December, 2nd 2020, again caused by a single household. The load of household #5 is increased significantly to about 2.5 kW from 5:22 p.m. until 7:06 a.m. Such behavior can indicate that an appliance was forgot to be turned off. This untypical demand pattern, barely visible in the aggregated load, is clearly recognizable in the prediction quantile and the anomaly score. The permanently high quantile indicates an unexpected high load.

In general, detecting anomalies that are created by just one household can be challenging using only the aggregated load. In this case, it was possible due to the high load and the moderate number of total households (21). The success of detecting such cases might decrease with more aggregated households or with higher load variability, e.g., during peak consumption periods in evenings.

Moreover, during the period from March to May in 2020, we find an increased number of situations with a high anomaly score. This can most likely be attributed to the lockdown in Germany due to the COVID-19 pandemic during this period.

D. Detection Performance with synthetic Anomalies

To evaluate AD performance comprehensively, we introduce synthetic anomalies. These anomalies include point anomalies with a single data point being 0 or unusually high, extraordinary over- or under-consumption, e.g. due to collective behavior, communication issues with erroneous measurement values, and a switching attack.

Fig. 4 showcases the application of our AD method for an excerpt of the testing period. The according quantile of the actual load in regard to the probabilistic forecast is shown. Despite certain fluctuations, the quantile clearly reaches extreme values during the anomalies, either because the load is extraordinarily high or low for a longer time. The resulting anomaly score (4) stays low during normal situations and rises during the anomaly periods, allowing for distinctive detection. As can be seen, all of the anomalies can be timely detected.

Table II summarizes classification metrics for the evaluation of a full year (2020) with 18 inserted anomalies. We use Recall, Precision, and F1 Score at the thresholds of maximized F1 Score, as well as the area under curve (AUC) of the ROC. We compare the proposed Quantile Score from (3) and Quantile Score Smoothed from (4) with state-of-the-art methods, like relative threshold and absolute threshold, the latter including the 2-sigma rule. In all metrics, the quantile-based methods outperform the deterministic ones.

To further investigate the characteristics of our detection method, we analyze the ROC. Fig. 5 shows the ROC curve for the same detection methods as in Table II. For the deterministic threshold methods, a considerable false positive rate (FPR) is

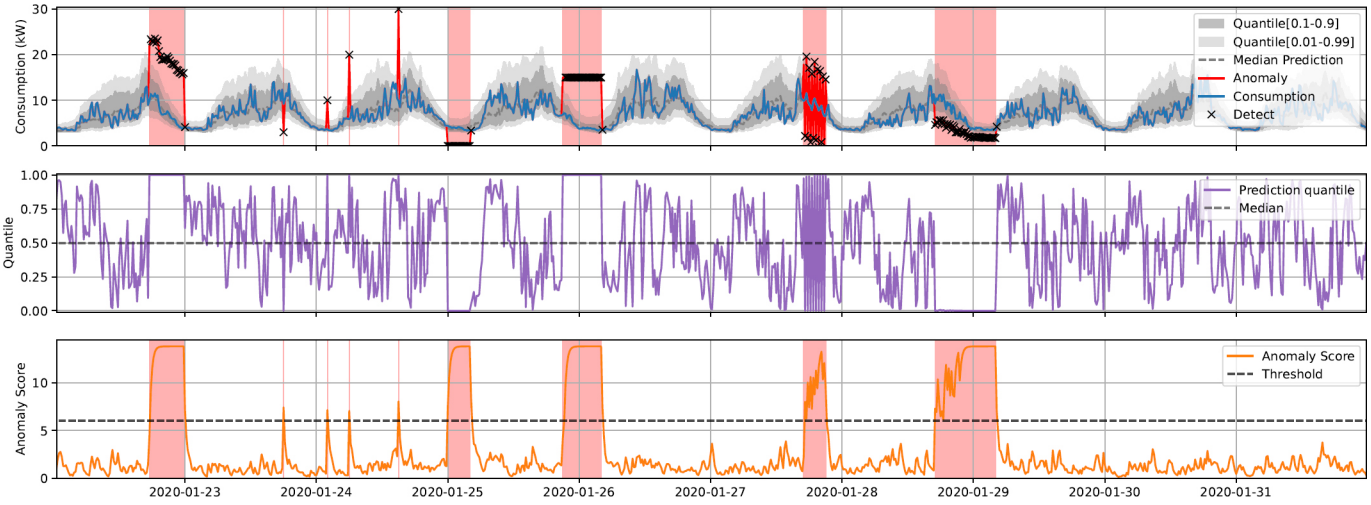


Fig. 4. Detection of synthetic anomalies with QR/ARMAX forecast, showing the actual load (blue) and the predicted load including confidence intervals (gray, dashed), the associated quantile of the prediction distribution, and the resulting anomaly score (smoothed).

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TABLE II
CLASSIFICATION PERFORMANCE OF DETECTION METHODS

Method	F1 Score	Recall	Precision	AUC
Quantile Regression (Auto-regr. + Ex. var.)				
Abs. Threshold	0.5268	0.4765	0.5891	0.95753
Rel. Threshold	0.4991	0.4235	0.6076	0.98222
Quantile Score	0.8796	0.9559	0.8145	0.99804
Qu. Score Smooth.	0.9277	0.9059	0.9506	0.99961
Quantile XGBoost (Auto-regr. + Ex. var.)				
Abs. Threshold	0.5097	0.4265	0.6332	0.95875
Rel. Threshold	0.5000	0.4206	0.6164	0.98346
Quantile Score	0.8946	0.9735	0.8275	0.99864
Qu. Score Smooth.	0.9467	0.9412	0.9524	0.99979
Quantile Regression (Ex. var. only)				
Abs. Threshold	0.4904	0.3765	0.7033	0.94423
Rel. Threshold	0.4538	0.3324	0.7152	0.96684
Quantile Score	0.9222	0.9412	0.9040	0.99797
Qu. Score Smooth.	0.9430	0.9000	0.9903	0.99903
Quantile XGBoost (Ex. var. only)				
Abs. Threshold	0.4910	0.4029	0.6284	0.94857
Rel. Threshold	0.4260	0.4147	0.4379	0.97533
Quantile Score	0.8991	0.9824	0.8288	0.99850
Qu. Score Smooth.	0.9466	0.9382	0.9551	0.99978

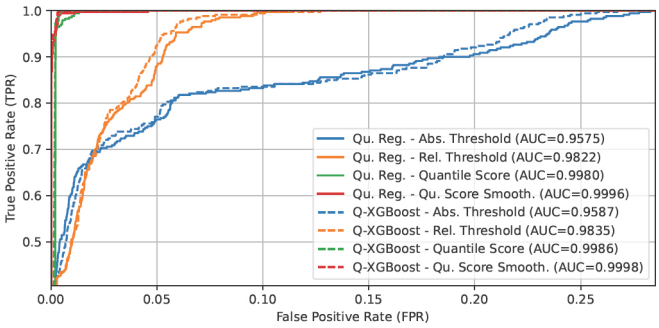


Fig. 5. ROC curve with different detection methods

needed for achieving a high true positive rate (TPR). The difference in the LF method is not significant.

V. DISCUSSION

The primary challenge of any anomaly detector is to differentiate between genuinely abnormal situations and usual demand fluctuations, which can be regarded as mere "noise". Monitoring of the active load is particularly challenging, due to the inherent stochasticity of electricity demand patterns, compared to e.g. heating systems [22]. The temporal resolution of the data might play a crucial role. Most previous studies considered hourly or daily data resolution, where the load appears more deterministic, as fluctuations average out, and deterministic detection methods may suffice. In contrast, we focus on real-time AD for fast detection, requiring very-short forecast horizons, where unpredictable fluctuations are present. Our evaluation on 15-minute data demonstrates that the probabilistic approach is well suited to handle the variability in the data effectively. Moreover, the number of households contributing to the load can affect detection: with few households, single household anomalies can be more easily identified, whereas with larger aggregates, the fluctuations smooth out, making collective deviations more easily detectable.

Comparing classification performance scores among different AD studies is problematic due to the use of varying datasets and evaluation metrics, hindering effective evaluation and overarching comparison of methods. In addition, the lack of labeled real-world anomaly data has forced many studies to artificially insert anomalies for evaluation [7], a practice that may skew the results. Even when real anomaly data is available, the context-specific nature of these events and the difficulty in defining clear boundaries between "normal" and "anomalous" conditions render binary classification assessments limited for practical applications.

Forecasting-based AD can be fully modularized, separating the LF model from the detection process, which is the AS module plus thresholding in our framework (see Fig. 2). As long as the interface (a load forecast for each time step) is maintained, this modular setup provides flexibility, since the LF or AS module can be independently replaced or optimized to enhance overall AD performance. While many studies pursuing the forecasting-based approach seem to focus primarily on improving the LF accuracy, especially median prediction, e.g. as MAPE, a superior forecast model does not necessarily translate into better AD performance. In our experiments, we saw that although XGBoost achieved higher forecast accuracy, the improvement in AD performance was negligible compared to the linear QR model. QR even yielded a better F1 score when used without auto-regressive variables, suggesting that LF overfitting can limit AD performance. In contrast to the growing trend of applying complex DL techniques for LF, our study shows that the probabilistic AS technique performs robustly even with less precise forecasts. This means that simpler, inherently explainable LF models such as linear ARMAX/QR can be used for AD, offering greater interpretability, while still achieving high detection performance.

Transparency is an important aspect for grid monitoring, as operators require explainable indicators to contextualize and respond to anomaly alerts. The proposed anomaly score in its transparent design can serve as an intuitive, continuous indicator for the degree of abnormality of the observed load in regard to the expected load. The anomaly score and the prediction quantile could be used for real-time monitoring by providing traceable insights, e.g., through visualization tools, making it accessible for operational use. Moreover, the anomaly score can be normalized, adapted to context-specific thresholds, or combined with additional indicators to create more comprehensive metrics, providing flexibility. This allows grid operators to integrate additional sources of anomaly signals into a unified early-warning system. The probabilistic model also enables the quantification of associated risks of system-critical situations, providing deeper context for risk management and decision-making under uncertainty.

AD methods should also consider generalizability for the transfer to new application contexts without sufficient data [14]. Most ML-based approaches (both supervised and unsupervised) rely on large datasets, which may result in overfitting, especially with intensive hyper-parameter tuning, making them less adaptable when facing new or diverse application environments.

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In contrast, our AS technique is defined in a statistically general way, not dependent on context-specific settings, allowing for application in entirely different contexts, e.g. as done in [15]. The only prerequisite of our approach is a sufficiently accurate probabilistic forecast. While a forecast for the target context may require limited historical "normal" data, the generalization of LF models is much better understood and manageable compared to purely data-driven AD methods, due to the domain-specific context. Depending on the method, the LF model itself is potentially susceptible to overfitting, e.g. with excessive feature complexity, but this risk can be mitigated with simple models like QR, which keeps each feature influence transparent.

Despite the promising application of PLF, we encountered technical difficulties of the PLF models, such as quantile crossing, which needs to be considered in the implementation. Also, it appears that the quantile-based regression cannot utilize extreme quantile values near 0 and 1, leading to certain upper and lower prediction bounds. To fully quantify the anomalousness of extreme spikes that exceed these thresholds, the AS module could be adjusted to incorporate measures of absolute exceedance beyond these bounds.

VI. Conclusion

This paper introduced a new forecasting-based AD approach. We focused on a statistical, explainable method and demonstrated that the probabilistic approach outperforms deterministic state-of-the-art methods. QR-based forecasting could compete with XGBoost as a non-linear model, indicating that linear forecast models are sufficient for achieving high detection performance, when combined with the proposed quantile-based anomaly score. The transparent design of our method allows for user interpretability, explainability and can assist grid operators in real-time monitoring.

Future work could test other PLF methods or AS techniques. Moreover, applications at even lower resolutions, e.g. 1 minute, might be feasible, as the probabilistic approach seems to be capable of dealing with the short-term fluctuations effectively.

In this study, we only considered the aggregated load of residential consumers as input. However, with the increasing roll-out of smart meters, the integration of individual real-time load data could be of interest for more granular AD.

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